SLIIT FACULTY OF COMPUTING | CDAP RESEARCH DEFENSE R26-IT-162

FloodSense AI: Intelligent IoT Microclimate Sensing & Dual-Stage Predictive Flood Inundation Modeling

Hyperlocal Rainfall Forecasting & Distance-Based Dynamic Risk Scoring for Kelani River Basin Communities

STUDENT A (IT21313998)
Rasara S.H.K. — Embedded IoT & Rainfall ML

- ESP32 calibrated tipping bucket rain gauge node (0.2mm per tip)
- HiveMQ Cloud MQTT broker & non-blocking signal processing
- Open-Meteo 14-variable atmospheric live data fusion engine
- LightGBM/XGBoost multi-horizon rainfall forecasting (24h, 48h, 72h)

STUDENT B (IT22236814)
U.H.K. Sankalana — Flood Risk & SaaS Dashboard

- Distance-Based Dynamic Flood Risk Personalization (DFRS)
- Stage 1A: XGBoost Binary Occurrence Classifier (F1 = 0.941)
- Stage 1B: XGBoost Continuous Risk Regressor (0–100 Scale)
- Stage 2: Multi-Output Regressor (Arrival Time hrs & Water Height m)
- React SaaS Dashboard with Sinhala, Tamil & English alerts

PROBLEM CONTEXT & SOCIO-ECONOMIC IMPACT

The Kelani River Basin Flood Crisis & Early Warning Deficit

500,000+ vulnerable citizens face severe recurring disaster across Colombo, Gampaha & Kegalle

Current Systemic Deficits

- Coarse 25 km synoptic grids miss micro-catchment cloudbursts.
- Manual gauge reporting collects water levels once every 6–12 hours.
- Binary blanket alerts lack doorstep personalization.
- No inundation arrival time or localized water depth forecasting.

Humanitarian & Economic Cost

- LKR 50+ Billion in direct annual economic losses.
- 300,000+ people displaced annually in urban lower reaches.
- 50–100+ fatalities in catastrophic monsoon events (2016, 2021).
- 15–30 days of recurring transport and business disruption.

REVIEWER FEEDBACK & SYSTEM EVOLUTION

How We Addressed Reviewer Comments & Upgraded the Research

Systematic progression from a basic monitoring prototype to an enterprise-grade AI flood early warning system

Sensor & Pipeline Evolution

• From: Single sensor monitoring ➔ To: Multi-factor IoT array (Tipping bucket, reed switch debouncing ISR, Open-Meteo fusion).

• From: Manual logs & once-a-day polling ➔ To: Sub-second MQTT streaming via HiveMQ Cloud Broker.

AI Modeling & Alerting Evolution

• From: Binary flood alerts (Yes/No) ➔ To: Dual-stage ML pipeline providing continuous risk score (0-100), arrival time, and depth.

• From: Generic text notifications ➔ To: Personalized Distance-Based Dynamic Risk (DFRS) in Sinhala, Tamil, English.

SYSTEM ARCHITECTURE & DATA FLOW

The Integrated Symbiotic Pipeline: Sensor to Inundation Decision

Seamless cross-tier data pipeline connecting Student A's IoT & ML engine with Student B's flood risk models

Component A: Sensing & Rain ML

- ESP32 Microcontroller with interrupt-driven tipping bucket.
- HiveMQ Cloud MQTT Broker (sub-second telemetry ingestion).
- Open-Meteo live atmospheric fusion (14 variables).
- LightGBM multi-horizon predictor for 24h, 48h, and 72h rainfall.

Component B: Two-Stage Flood ML & UI

- Stage 1A XGBoost Binary Classifier (Occurrence Yes/No).
- Stage 1B XGBoost Regressor (Continuous Risk Score 0–100).
- Stage 2 Multi-Output Regressor (Arrival Time & Water Height).
- Vite React Dashboard with trilingual emergency SOP notifications.

COMPONENT A | RASARA S.H.K. (IT21313998)

Embedded IoT Microclimate Sensing & Edge Telemetry Firmware

High-frequency rainfall pulse detection with hardware debouncing and resilient cloud messaging

Hardware Architecture

- ESP32 dual-core Xtensa 32-bit LX6 running at 240 MHz.
- Tipping bucket rain collector (calibrated at 0.20 mm per tip).
- DHT22 / BME280 temperature and relative humidity monitoring.
- Solar-powered off-grid subsystem (3.7V Li-ion + 5W solar + TP4056).

Signal Processing & MQTT

- Interrupt Service Routine (ISR) executed in high-speed IRAM.
- 100 ms software debounce lock eliminating reed contact chatter.
- Median filter rejecting false positive spikes from wind-blown debris.
- Lightweight MQTT v3.1.1 telemetry over HiveMQ Cloud Broker.

COMPONENT A | RASARA S.H.K. (IT21313998)

Multi-Horizon Machine Learning Rainfall Forecasting Engine

Coupling atmospheric reanalysis with edge telemetry to predict 24h, 48h, and 72h precipitation depths

Data Fusion & Feature Pipeline

- 14 Open-Meteo variables (CAPE, 2m temp, surface pressure, RH, cloud).
- Orthogonal wind vector decomposition: u = -s·sin(θ) and v = -s·cos(θ).
- Rolling antecedent windows (1h, 3h, 6h, 12h, 24h, 48h, 72h, 7-day).
- Antecedent Precipitation Index (API) with recession constant k = 0.88.

Model Benchmarks & Results

- ■ LightGBM Regressor: R^2 = 0.841 | MAE = 2.84 mm | RMSE = 4.12 mm.
- ■ XGBoost Regressor: R^2 = 0.835 | MAE = 2.98 mm | RMSE = 4.25 mm.
- Rain Occurrence Classifier: F1-score = 0.912 and ROC-AUC = 0.945.
- Zero lookahead bias enforced via TimeSeriesSplit cross-validation.

COMPONENT B | U.H.K. SANKALANA (IT22236814)

Two-Stage Machine Learning Dynamic Flood Risk & Inundation Model

Combining hydro-topographic vectors with rainfall predictions to compute occurrence, risk score, arrival time, and depth

Stage 1A & 1B Models

- Stage 1A: XGBoost Classifier (F1 = 0.941) evaluates Flood Occurrence.
- Bayesian posterior probability calibrated against 10-year Kelani records.
- Stage 1B: XGBoost Regressor (RMSE = 3.12) calculates 0–100 Risk Score.
- Categorized into Low (<30), Moderate (30–49), High (50–69), Very High (≥70).

Stage 2 Multi-Output Model

- Executes conditionally when Stage 1A predicts Flood Occurrence = Yes.
- Metric 1: Inundation Arrival Time (hours until wavefront reaches community).
- Metric 2: Peak Inundation Height (water depth in meters above ground).
- Integrates distance to river, terrain elevation, and soil classification.

COMPONENT B | U.H.K. SANKALANA (IT22236814)

Distance-Based Dynamic Flood Risk Personalization (DFRS)

Hyperlocal spatial proximity modeling, executive React dashboard UI, and trilingual community alerts

Distance-Based Personalization

- Proximity decay algorithm scales risk as river channel distance decreases.
- Couples digital elevation (DEM/SRTM), soil type (Clay), and land cover.
- Sedawatta (45m river distance, 3.2m elevation) ➔ High Risk (Arrival 3.4h, Height 3.84m).
- Malabe (1,250m river distance, 32m elevation) ➔ Safe baseline.

Executive React Dashboard

- Tailwind CSS with SVG semi-circular risk dials and model KPI banners.
- Live API health check indicator monitoring Flask backend on port 5000.
- 1-Click Viva Demonstration Presets (Sedawatta, Ja-Ela, Malabe).
- Trilingual notification dispatch in Sinhala, Tamil, and English.

CROSS-SYSTEM INTEGRATION & DATA CONTRACT

How Component A Feeds Component B in Real-Time

The automated mapping pipeline transforming multi-horizon rainfall forecasts into the 12-feature flood matrix

Component A Handoff Payload

- rainfall_7d_mm: Cumulative antecedent precipitation (e.g. 200 mm).
- monthly_rainfall_mm: 30-day baseline catchment saturation (e.g. 350 mm).
- rainfall_intensity_mmhr: Peak storm cloudburst intensity (e.g. 15–28 mm/hr).
- Exposed via automated REST API (/api/v1/forecast/latest) and in-process bridge.

Component B Execution Result

- Binds Component A's rainfall with 9 geospatial features (lat/lon, elevation, soil).
- Clicking '■ Sync IoT Rainfall' in React UI executes live prediction.
- Yields: Occurrence = YES, Risk = 56.92%, Arrival = 3.41h, Water Height = 3.84m.
- Demonstrates genuine end-to-end multi-disciplinary research cohesion.

EMPIRICAL VALIDATION & COMMUNITY FIELD TRIALS

Field Testing Across Kelani Riverine Communities & Post-Test Iterations

Tested over 3 weeks across 3 agricultural riverine locations capturing 12 real monsoon rainfall events

Field Trial Protocol & Scores

- Tested across Sedawatta (Lower Reach), Hanwella, and Kaduwela.
- 15 riverine households, 5 farmers, and 2 disaster coordinators participated.
- Overall user satisfaction rating: 4.4 / 5.0.
- Alert timeliness: 4.6 / 5.0 (average 4.2 hours actionable evacuation lead time).

Post-Testing Iterations Implemented

- Added 5-sample median filtering in firmware to prevent false tips from debris.
- Configurable distance weighting added to DFRS personalization algorithm.
- Optimized deep-sleep power cycling, extending off-grid battery life to 90 days.
- Automated SMS receipt acknowledgement added for emergency coordinators.

COMMERCIALIZATION & MARKET FEASIBILITY

Business Model, Unit Economics & 3-Year Financial Sustainability

A high-margin, scalable B2G & B2B SaaS model providing 26x cost disruption over imported stations

Multi-Stream Revenue Model

- B2G Municipal Contracts: Annual monitoring SLA with Disaster Management Centre (DMC) & Irrigation Dept (LKR 1.5M - 2.5M/yr).
- Insurance Data-as-a-Service (DaaS): Parametric flood risk verification API for insurers (Ceylinco, Sri Lanka Insurance) at LKR 250k/mo.
- B2B Commercial Agriculture: Real-time microclimate soil-moisture feeds for tea/rubber estates in upstream catchment reaches.

Unit Economics & Projections

- Per-Node CapEx: ~LKR 14,500 ($48 USD) vs Imported Stations LKR 400,000+ (26x Cheaper!).
- Pilot Phase (Year 1 - 50 Nodes): CapEx LKR 725k, Break-even in Month 8, Year 1 ROI ~87.5%.
- Growth Phase (Year 2 - 150 Nodes): Revenue LKR 5.8M with 48% net margin.
- Scale Phase (Year 3 - 450 Nodes National): Revenue LKR 14.5M across 4 major river basins.

DEPLOYMENT ROADMAP & SOCIAL IMPACT

Phased National Rollout Plan & Sustainable Community Resilience

From Kelani river basin pilot to a nationwide disaster early warning network protecting 11M+ citizens

Phased Rollout Strategy

- Phase 1 (Year 0–1): 50-node Kelani Basin pilot, DMC integration, 500,000 people protected.
- Phase 2 (Year 2): Scale to 150 nodes covering entire Kelani basin; onboard insurance partners.
- Phase 3 (Year 3–5): National rollout to Kalu Ganga, Gin Ganga, and Nilwala basins (450+ nodes).

Social Mission & Sustainability

- 100% free SMS warnings for registered low-income riverine households.
- Solar-powered off-grid hardware with local youth technician maintenance training.
- Supports UN Sustainable Development Goals (SDG 11: Sustainable Cities, SDG 13: Climate Action).
- Partnership pathways with World Bank and Asian Development Bank Climate Facilities.

CONCLUSION & TRANSITION TO DEMONSTRATION

Key Research Contributions & Live Demonstration Sequence

Transforming theoretical machine learning into an actionable, life-saving early warning system

Research Contributions

- Sub-kilometer spatial disruption solving the 25km synoptic forecasting blind-spot.
- Multi-horizon LightGBM predictive framework ($R^2 = 0.841$, MAE = 2.84 mm).
- Two-Stage Inundation Intelligence predicting arrival time and water depth.
- 26x cost innovation proving sustainable commercial viability for Sri Lanka.

Live Demonstration Sequence

- 1. Live IoT Telemetry: ESP32 node streaming pulses to HiveMQ Cloud.
- 2. AI Weather Fusion: FastAPI backend fetching live Open-Meteo grids.
- 3. AI Forecast Dashboard: Streamlit multi-horizon predictive curves.
- 4. Storm Simulation Mode: Injecting 250mm cloudburst scenario.
- 5. End-to-End React UI Sync: Executing Two-Stage XGBoost model live.